Title: Enterprise SSH Defense & SIEM Detection Project
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Enterprise SSH Defense & SIEM Detection

Objective

To simulate enterprise-level SSH attacks and implement centralized log monitoring using a SIEM setup to detect, analyze, and respond to security events.

Tools Used

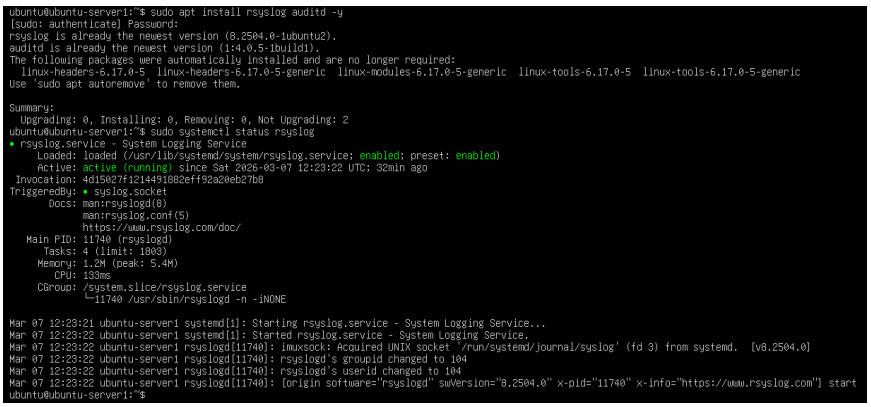
- rsyslog (log forwarding)
- auditd (system auditing)
- Hydra (brute-force attack)
- SSH
- Ubuntu Server
- Kali Linux

Lab Architecture

- Attacker: Kali Linux
- Victim: Ubuntu Server
- SIEM: Ubuntu Server (log collector)
- Logs are forwarded from victim → SIEM using rsyslog (TCP 514)

Log Collection Setup

rsyslog and auditd were installed on both victim and SIEM servers to enable system logging and activity monitoring.



```
ubuntu@ubuntu-server1:~$ sudo apt install rsyslog auditd -y
[sudo: authenticate] Password:
rsyslog is already the newest version (8.2504.0-1ubuntu2).
auditd is already the newest version (1:4.0.5-1build1).
The following packages were automatically installed and are no longer required:
  linux-headers-6.17.0-5  linux-headers-6.17.0-5-generic  linux-modules-6.17.0-5-generic  linux-tools-6.17.0-5  linux-tools-6.17.0-5-generic
Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 2
ubuntu@ubuntu-server1:~$ sudo systemctl status rsyslog
● rsyslog.service - System Logging Service
     Loaded: loaded (/usr/lib/systemd/system/rsyslog.service; enabled; preset: enabled)
     Active: active (running) since Sat 2026-03-07 12:23:22 UTC; 32min ago
 Invocation: 4d15027f121449108 2eff92a20eb27b8
TriggeredBy: ● syslog.socket
       Docs: man:rsyslogd(8)
             man:rsyslog.conf(5)
             https://www.rsyslog.com/doc/
   Main PID: 11740 (rsyslogd)
      Tasks: 4 (limit: 1003)
     Memory: 1.2M (peak: 5.4M)
        CPU: 133ms
     CGroup: /system.slice/rsyslog.service
             └─11740 /usr/sbin/rsyslogd -n -iNONE

Mar 07 12:23:21 ubuntu-server1 systemd[1]: Starting rsyslog.service - System Logging Service...
Mar 07 12:23:22 ubuntu-server1 systemd[1]: Started rsyslog.service - System Logging Service.
Mar 07 12:23:22 ubuntu-server1 rsyslogd[11740]: imuxsock: Acquired UNIX socket '/run/systemd/journal/syslog' (fd 3) from systemd.  [v8.2504.0]
Mar 07 12:23:22 ubuntu-server1 rsyslogd[11740]: rsyslogd's groupid changed to 104
Mar 07 12:23:22 ubuntu-server1 rsyslogd[11740]: rsyslogd's userid changed to 104
Mar 07 12:23:22 ubuntu-server1 rsyslogd[11740]: [origin software="rsyslogd" swVersion="8.2504.0" x-pid="11740" x-info="https://www.rsyslog.com"] start
ubuntu@ubuntu-server1:~$
```

Figure 1: rsyslog service running and confirmed active on the victim server.

```
ubuntu@ubuntu-server1:~$ sudo systemctl status auditd
• auditd.service - Security Audit Logging Service
     Loaded: loaded (/usr/lib/systemd/system/auditd.service; enabled; preset: enabled)
     Active: active (running) since Sat 2026-03-07 12:24:59 UTC; 1h 26min ago
 Invocation: 78b9b5bafe444d719b45c933cd3256ed
       Docs: man:auditd(8)
             https://github.com/linux-audit/audit-documentation
    Process: 12097 ExecStart=/usr/sbin/auditd (code=exited, status=0/SUCCESS)
   Main PID: 12107 (auditd)
      Tasks: 2 (limit: 1803)
     Memory: 528K (peak: 1.7M)
        CPU: 36ms
     CGroup: /system.slice/auditd.service
             └─12107 /usr/sbin/auditd

Mar 07 12:24:59 ubuntu-server1 systemd[1]: Starting auditd.service - Security Audit Logging Service...
Mar 07 12:24:59 ubuntu-server1 auditd[12107]: No plugins found, not dispatching events
Mar 07 12:24:59 ubuntu-server1 auditd[12107]: Init complete, auditd 4.0.5 listening for events (startup state enable
Mar 07 12:24:59 ubuntu-server1 systemd[1]: Started auditd.service - Security Audit Logging Service.
ubuntu@ubuntu-server1:~$
```

Figure 2: auditd service running and active on the victim server.

```
siem@ubuntu-siem:~$ sudo apt install rsyslog auditd -y
[sudo] password for siem:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
rsyslog is already the newest version (8.2312.0-3ubuntu9.1).
auditd is already the newest version (1:3.1.2-2.1build1.1).
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
siem@ubuntu-siem:~$ sudo systemctl status rsyslog
• rsyslog.service - System Logging Service
     Loaded: loaded (/usr/lib/systemd/system/rsyslog.service; enabled; preset: enabled)
     Active: active (running) since Sat 2026-03-07 12:20:31 UTC; 37min ago
TriggeredBy: • syslog.socket
       Docs: man:rsyslogd(8)
             man:rsyslog.conf(5)
             https://www.rsyslog.com/doc/
   Main PID: 2113 (rsyslogd)
      Tasks: 4 (limit: 2205)
     Memory: 1.3M (peak: 5.3M)
        CPU: 90ms
     CGroup: /system.slice/rsyslog.service
             └─2113 /usr/sbin/rsyslogd -n -iNONE

Mar 07 12:20:31 ubuntu-siem systemd[1]: Starting rsyslog.service - System Logging Service...
Mar 07 12:20:31 ubuntu-siem rsyslogd[2113]: imuxsock: Acquired UNIX socket '/run/systemd/journal/sysl
Mar 07 12:20:31 ubuntu-siem rsyslogd[2113]: rsyslogd's groupid changed to 104
Mar 07 12:20:31 ubuntu-siem systemd[1]: Started rsyslog.service - System Logging Service.
Mar 07 12:20:31 ubuntu-siem rsyslogd[2113]: rsyslogd's userid changed to 103
Mar 07 12:20:31 ubuntu-siem rsyslogd[2113]: [origin software="rsyslogd" swVersion="8.2312.0" x-pid="2
siem@ubuntu-siem:~$ _
```

Figure 3: rsyslog service running on the SIEM server for centralized log collection.

Log Forwarding

The victim server was configured to forward logs to the SIEM server using rsyslog over TCP port 514. This enables centralized monitoring of security events.

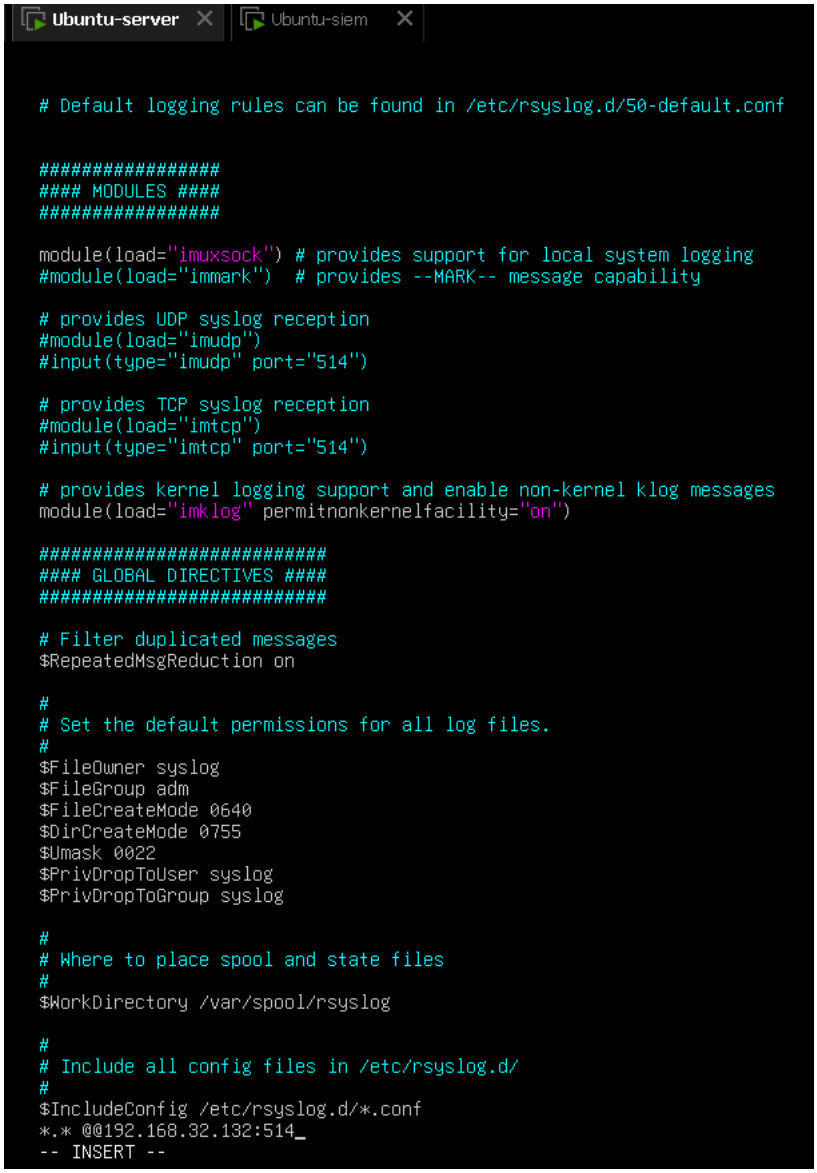


Figure 4: Configuring rsyslog on the victim server to forward logs to SIEM server.

```
## First rule - delete all
-D

## Increase the buffers to survive stress events.
## Make this bigger for busy systems
-b 8192

## This determine how long to wait in burst of events
--backlog_wait_time 60000

## Set failure mode to syslog
-f 1

-w /etc/passwd -p wa -k passwd_changes
-w /etc/shadow -p wa -k shadow_changes
-w /etc/ssh/sshd_config -p wa -k sshd_config_changes
-w /home -p wa -k home_changes
~
~
```

Figure 5: Adding auditd rules on the victim server to monitor security-related events.

```
# /etc/rsyslog.conf configuration file for rsyslog
#
# For more information install rsyslog-doc and see
# /usr/share/doc/rsyslog-doc/html/configuration/index.html
#
# Default logging rules can be found in /etc/rsyslog.d/50-default.conf


#################
#### MODULES ####
#################

module(load="imuxsock") # provides support for local system logging
#module(load="immark")  # provides --MARK-- message capability

# provides UDP syslog reception
module(load="imudp")
input(type="imudp" port="514")

# provides TCP syslog reception
module(load="imtcp")
input(type="imtcp" port="514")

# provides kernel logging support and enable non-kernel klog messages
module(load="imklog" permitnonkernelfacility="on")

###########################
#### GLOBAL DIRECTIVES ####
###########################

# Filter duplicated messages
$RepeatedMsgReduction on

#
# Set the default permissions for all log files.
#
$FileOwner syslog
$FileGroup adm
$FileCreateMode 0640
$DirCreateMode 0755
$Umask 0022
$PrivDropToUser syslog
$PrivDropToGroup syslog

#
# Where to place spool and state files
#
$WorkDirectory /var/spool/rsyslog

-- INSERT --
```

Figure 6: rsyslog configuration on the SIEM server to receive and process incoming logs.

Attack Simulation

A brute-force attack was performed using Hydra from the attacker machine targeting SSH service on the victim server.

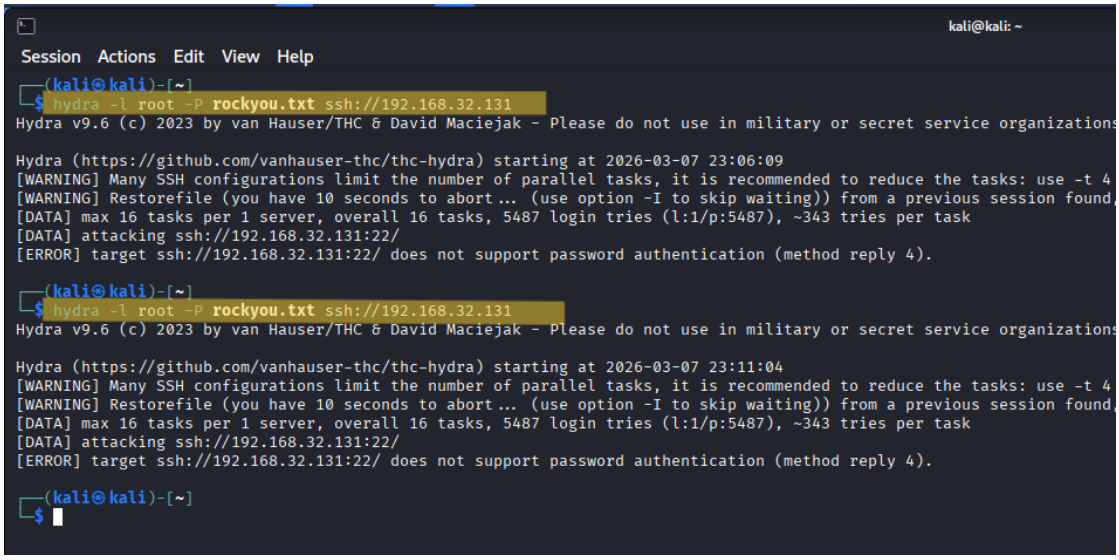


Figure 7: Brute-force attack initiated from attacker machine targeting SSH service.

Detection & Analysis

- The SIEM server successfully received logs related to failed SSH login attempts, allowing detection of brute-force activity in real time.
- This demonstrates the effectiveness of centralized logging in identifying suspicious authentication activity in real time.

```
siem@ubuntu-siem:~$ sudo tail -f /var/log/auth.log
[sudo] password for siem:
2026-03-07T17:25:01+00:00 ubuntu-server1 CRON[13123]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
2026-03-07T17:25:01+00:00 ubuntu-server1 CRON[13123]: pam_unix(cron:session): session closed for user root
2026-03-07T17:35:01.642711+00:00 ubuntu-siem CRON[3563]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
2026-03-07T17:35:01.646178+00:00 ubuntu-siem CRON[3563]: pam_unix(cron:session): session closed for user root
2026-03-07T17:35:01+00:00 ubuntu-server1 CRON[13137]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
2026-03-07T17:35:01+00:00 ubuntu-server1 CRON[13137]: pam_unix(cron:session): session closed for user root
2026-03-07T17:36:19+00:00 ubuntu-server1 sshd-session[13142]: Received disconnect from 192.168.32.130 port 40988:11: Bye Bye [preauth]
2026-03-07T17:36:19+00:00 ubuntu-server1 sshd-session[13142]: Disconnected from authenticating user root 192.168.32.130 port 40988 [preauth]
2026-03-07T17:39:49.398831+00:00 ubuntu-siem sudo:     siem : TTY=tty1 ; PWD=/home/siem ; USER=root ; COMMAND=/usr/bin/tail -f /var/log/auth.log
2026-03-07T17:39:49.403478+00:00 ubuntu-siem sudo: pam_unix(sudo:session): session opened for user root(uid=0) by siem(uid=1000)
2026-03-07T17:41:15+00:00 ubuntu-server1 sshd-session[13151]: Received disconnect from 192.168.32.130 port 48082:11: Bye Bye [preauth]
2026-03-07T17:41:15+00:00 ubuntu-server1 sshd-session[13151]: Disconnected from authenticating user root 192.168.32.130 port 48082 [preauth]
```

Figure 8: SSH brute-force login attempts successfully detected and logged on the SIEM server.

Persistence Attack Simulation

- A persistence mechanism was simulated by adding an unauthorized SSH key to the authorized_keys file on the victim server.
- This allowed passwordless access from the attacker machine.
- Log tampering was also simulated by stopping rsyslog to evade detection.
- This technique simulates how attackers maintain long-term access without relying on passwords.

```
Ubuntu-server    Ubuntu-siem

ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAACAQCbu2ieUPQRzt3QOJSL/gta9r7sx/6P7jMx8TUsNxq4pAljg8kGWXR8rHeymINK9jP9+jZq1PE3H6s3eqy6yjxhe/oZtIni2CdYZmp3p8tmApFuCwnqUMp/DJaK
/xP1/kisHyRTBf81YLLLKM+bcphwsvghsQKqDV4BhyS8+dKIp6lzlmZ9Eh3nmVlIjxWgtml6fyx43jA8jRfys1E0xHk1maMd1JFVd+huwSY7e0emimd74VZXppkYALgHS/kV7z6+BUBSLAuSQvYCbu/+xO3bFvUf
A1nUF8xQhylu8vN5TtRV0hAUW/zcqf2lgR0Ge8rFSiOJWCLECu2X9LaYIloWGJJovmiTv/22XjCmSqQndtUWsPBmK4O5lNpNN0RWuNlQSp2nJGXfVeGnlKJKvsp/9nwk3BhbBkG85xl2saK9/XgtMTCKCVeOO2qf
Ia17kVQ2EWuiS55/ALsMYsYJOfocv2aRNWyVxcM6awVOsqMI5BsATHHxvVk5iqF1WBGaIUdzmnn2YGR7yjtsSvtV2vnTxL/nsfpv4GRzOVZgJAqvT7rqtxn/CEwMqGvNOTFrgozv0bmGYN+DZY0cZf4Ad564n
JrpFAJRo6x7GR9GpXc8ii8wKgH00F3id1GCEt+Rqyh4nANVRscvVvm2ZS0FH2GnVC7TzlqZzM5EM1P0yfQ== kali@kali

-- INSERT --                                                                    1,735         All
```

Figure 9: Unauthorized modification of SSH authorized_keys file to establish persistence.

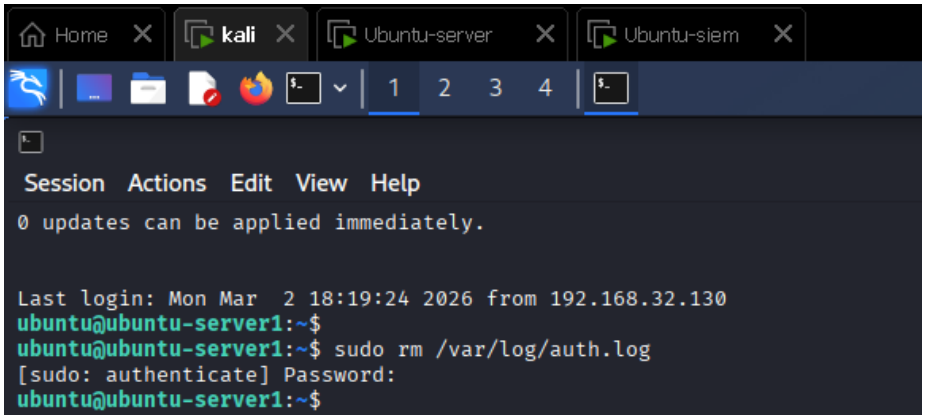


Figure 10: Log deletion attempt performed via SSH to evade detection.

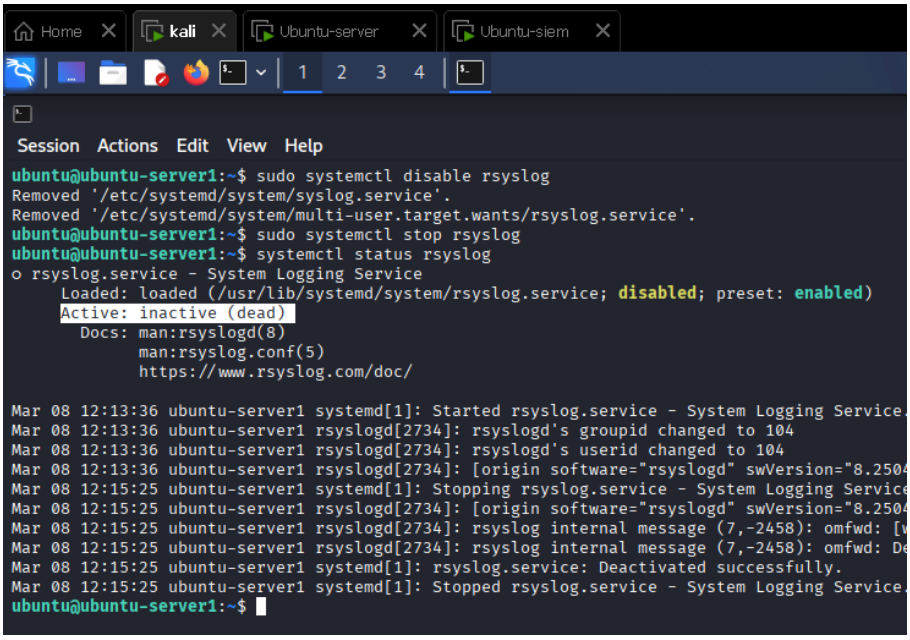


Figure 11: rsyslog stopped via SSH to disrupt log generation.

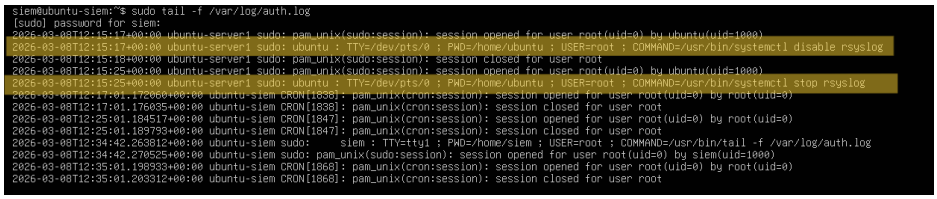


Figure 12: SIEM logs indicating termination of rsyslog service and absence of further log activity.

Results

- Centralized logging successfully implemented
- Brute-force attack detected via SIEM logs
- Unauthorized SSH key persistence identified
- Log disruption activity observed
- All attack activities were successfully captured and analyzed through centralized logging.

Key Learnings

- Importance of centralized log monitoring
- Detection of brute-force attacks through log analysis
- Risks of SSH key-based persistence
- Importance of log integrity and monitoring

Security Insights

- Centralized logging is critical for detecting distributed attacks
- Attackers may attempt log evasion by disabling logging services
- SSH key-based persistence is harder to detect than password attacks

Conclusion

This project demonstrates how centralized logging and monitoring can detect malicious activities such as brute-force attacks and persistence mechanisms, highlighting the importance of SIEM systems in enterprise security.

Appendix: Commands Used

1. Install Logging Packages
    - On Victim Server & SIEM Server
        - `sudo apt update`
        - `sudo apt install rsyslog auditd -y`

2. Enable & Start Services
    - `sudo systemctl enable rsyslog`
    - `sudo systemctl start rsyslog`

    - `sudo systemctl enable auditd`
    - `sudo systemctl start auditd`

3. Configure rsyslog on Victim
    - Edit rsyslog configuration:
        - `sudo vim /etc/rsyslog.conf`
    - Add this line at the end:
        - `*.* @@192.168.32.132:514`
    - Restart service:
        - `sudo systemctl restart rsyslog`

4. Configure rsyslog on SIEM Server:
    - Edit configuration:
        - `sudo vim /etc/rsyslog.conf`
    - Uncomment/add:
        - `module(load="imtcp")`
        - `input(type="imtcp" port="514")`
    - Restart rsyslog:
        - `sudo systemctl restart rsyslog`

5. Verify Log Reception (SIEM)
    - `sudo tail -f /var/log/syslog`

6. Configure auditd Rules (Victim)
    - Edit rules file:
        - `sudo vim /etc/audit/rules.d/audit.rules`

- Add:
  - `-w /etc/passwd -p wa -k passwd_changes`
  - `-w /etc/shadow -p wa -k shadow_changes`
  - `-w /etc/ssh/sshd_config -p wa -k ssh_config_changes`
  - `-w /home -p wa -k home_changes`
- Restart auditd:
  - `sudo systemctl restart auditd`

7. Brute-Force Attack (Kali Linux)
   - `hydra -l ubuntu -P /usr/share/wordlists/rockyou.txt ssh://192.168.32.131`

8. Persistence Attack (Add SSH Key)
   - On Kali (generate key)
     - `ssh-keygen`
   - Copy public key manually to victim:
     - `vim ~/.ssh/authorized_keys`
     - `(paste attacker public key)`

9. Log Deletion (Attack Simulation)
   - `sudo rm -rf /var/log/*`

10. Stop Logging Service (Evasion)
    - `sudo systemctl stop rsyslog`

11. Check Service Status
    - `sudo systemctl status rsyslog`
    - `sudo systemctl status auditd`